How-To Library

Informatica

Deploying the Informatica® Enterprise Data Preparation 10.4.0 Microsoft Azure Marketplace Solution

© Copyright Informatica LLC 2020, 2022. Informatica and the Informatica logo are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html.

Abstract

This deployment reference guide provides step-by-step instructions for deploying the Informatica® Enterprise Data Preparation 10.4.0 Microsoft Azure Marketplace Solution on the Microsoft Azure Marketplace. Automated reference deployments use Azure templates to launch, configure, and run the Azure compute, network, storage, and other services required to deploy a specific workload on Azure.

Supported Versions

- Enterprise Data Preparation 10.4.0

Table of Contents

Overview. 2
Intended Audience. 3
Costs and Licenses. 3
Architecture. 3
Azure Platform Resources in the Deployment. 5
Informatica Domain. 6
Informatica Clients. 7
Before You Begin. 8
License Key Prerequisite. 8
Prerequisites. 8
Deploying Enterprise Data Preparation on the Azure Marketplace. 9
Step 1. Begin Provisioning. 9
Step 2. Deploy Enterprise Data Preparation and Configure Azure Resources. 9
Step 3. Recycle the Informatica Services. 12
Monitoring Instance Provision and Informatica Domain Creation. 13
Logs. 14

Overview

Customers of Microsoft Azure and Informatica can execute an Enterprise Data Preparation deployment from the Azure marketplace to create an Informatica domain in the Azure and explore Enterprise Data Preparation functionality.

Enterprise Data Preparation is a collaborative self-service data discovery and preparation solution for data analysts and data scientists. It enables analysts to rapidly discover and turn raw data into insights and allows IT to ensure quality, visibility, and governance. With Enterprise Data Preparation, analysts to spend more time on analysis and less time on finding and preparing data.

This deployment reference guide provides step-by-step instructions for deploying Enterprise Data Preparation on the Microsoft Azure Marketplace. Automated reference deployments use Azure Resource Manager to launch, configure, and run the Azure compute, network, storage, and other services required to deploy a specific workload on Microsoft Azure.

Intended Audience

This guide is for users who are responsible for deploying the marketplace solution of Enterprise Data Preparation 10.4.0 on the Microsoft Azure Marketplace.

As a user with administrator privileges to deploy applications on Microsoft Azure, you must be familiar with Azure platform elements such as Azure Resource Manager, Virtual Machine, Virtual Network, Azure v2 storage accounts, Azure Active Directory, Azure HDInsight Clusters, Azure Blob storage, and Azure user managed identities.

You must also be familiar with Enterprise Data Preparation. To find the product documentation, see the Informatica documentation portal.

Costs and Licenses

You are responsible for the cost of the Azure services used while running the marketplace deployment. There is no additional cost for using the deployment.

The Azure resource manager template for this deployment includes configuration parameters that you can customize. Some of these settings, such as instance type, will affect the cost of deployment. See the pricing pages for each Azure service that you plan to use for cost estimates.

This deployment requires a license for Informatica Enterprise Data Preparation. To sign up for a demo license, contact your organization's Informatica sales contact or Informatica Global Customer Support.

Note: You supply the license key value in the Informatica License Key parameter when you configure the deployment.

The following table lists the instance types that you can choose based on sizing requirements:

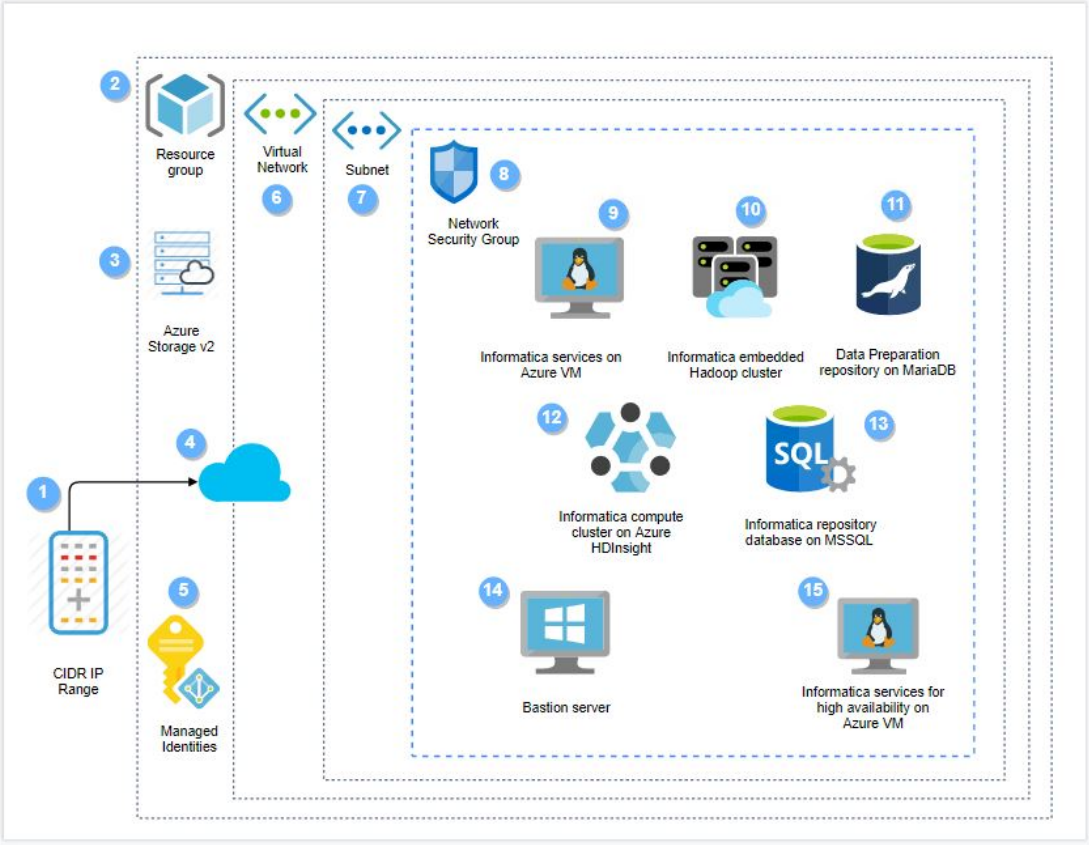
Virtual Machine	Instance Type	Cluster Size
Database	Standard_D3_v2 / Standard_DS4_v2 / Standard_DS11 / Standard_DS11_v2	Small, Medium, Large
Informatica Domain	Standard_D4_v2 / Standard_DS4_v2 / Standard_DS11 / Standard_DS11_v2	Small, Medium, Large
Bastion server	Standard_D2_v2/Standard_A2_v2/ Standard_DS2_v2/Standard_B2s	Small, Medium, Large
Informatica Cluster Service	Standard_D5_v2/Standard_DS5_v2/ Standard_E16_v3 Standard_D4_v2/Standard_DS4_v2/Standard_E16_v3	Small Medium, Large

Architecture

When you deploy the Enterprise Data Preparation marketplace solution on a Virtual Network, Microsoft Azure creates and connects the following resources in the network:

- An Informatica domain server on an Azure Virtual Machine, with one additional virtual machine for high availability.
- Informatica clients on a bastion server.
- A Microsoft SQL Server database for the Informatica repositories.
- A MariaDB database for the Data Preparation repository.
- An embedded Hadoop cluster on an Azure Virtual Machine instance.
- An HDInsight cluster.

The following image shows the architecture of the Informatica Enterprise Data Preparation 10.4.0 Microsoft Azure Marketplace Solution:



1. CIDR IP address range.
2. A resource group on the Azure platform.
3. Azure storage.
4. The Azure cloud.
5. Managed identities for the Azure resources.
6. A Virtual Network.
7. A subnet to contain the deployment.
8. A network security group.
9. The Informatica services on an Azure Virtual Machine instance.
10. An embedded Hadoop cluster running on HDInsight that Enterprise Data Catalog uses to run metadata processing and profiling jobs.
11. A MariaDB database for the Data Preparation repository.
12. A compute cluster running on HDInsight that Enterprise Data Preparation uses to publish prepared data to the data lake.
13. A Microsoft SQL Server database to contain the following Informatica repositories:
    - Domain configuration repository.
    - Model repository.
14. A bastion server, if you choose to deploy one.

15. Informatica services for high availability on an Azure Virtual Machine.

Azure Platform Resources in the Deployment

The deployment process creates or includes the following components:

- An Azure Virtual Machine for Informatica services and one for high availability.
- A Microsoft SQL Server database.
- A bastion server.
- An HDInsight cluster.
- A MariaDB database.

Components in a Deployment on a New Network

The following components are created when you deploy the software:

Component	Number of Components Created
Virtual Network	1
Network security group	1
Subnet	1
Microsoft SQL Server	1
Virtual machines	9: - 1 for Informatica domain - 1 for high availability - Up to 6 for the Informatica embedded Hadoop cluster - 1 for bastion server
HDInsight cluster nodes	- 2 for head node - 1, 3, or 6 worker nodes
MariaDB database	1
Azure Data Lake Storage Gen2 account	1
User-assigned managed identity	1

Components in a Deployment on an Existing Network

The following components are created when you deploy the software:

Component	Number of Components Created
Network security group	1
Microsoft SQL Server	1

Component	Number of Components Created
Virtual machines	3: - 1 for Informatica domain - 1 for high availability - 1 for bastion server Up to 7: - Up to 6 for Informatica depending on whether high availability is enabled - 1 for bastion server - 1 for Informatica domain - 1 for high availability - 1 for bastion server
MariaDB database	1

Bastion Server

You can optionally deploy a bastion server as an element in the solution. The bastion server acts as a firewall between the internet and the cloud platform network where you deploy the solution. It can also act as a remote server which you can log into to run Informatica clients.

Informatica Domain

The Informatica domain is a server component that hosts application services, such as the Model Repository Service and the Data Integration Service. These services, together with domain clients, enable you to create and run mappings and other objects to extract, transform, and write data.

Application Services

The Informatica domain includes the following application services:

Enterprise Data Preparation Service

The Enterprise Data Preparation Service is an application service in the Informatica domain that runs the Enterprise Data Preparation application.

Interactive Data Preparation Service

The Interactive Data Preparation Service is an application service in the Informatica domain that manages data preparation within the Enterprise Data Preparation application.

Catalog Service

The Catalog Service is an application service in the Informatica domain that runs the Enterprise Data Catalog application, and manages connections between service components and external applications.

Model Repository Service

The Model Repository Service is an application service in the Informatica domain that manages the Model repository. The Model repository stores metadata created by Informatica products in a relational database to enable collaboration among the products. Informatica Developer, the Data Integration Service, and the Administrator tool store metadata in the Model repository.

Data Integration Service

The Data Integration Service is an application service in the Informatica domain that performs data integration tasks for the Developer tool and for external clients.

Metadata Access Service

The Metadata Access Service is an application service that allows the Developer tool to access Hadoop connection information to import and preview metadata. The Metadata Access Service contains information about the Service Principal Name (SPN) and keytab information if the Hadoop cluster uses Kerberos authentication.

Content Management Service

The Content Management Service is an application service in the Informatica domain that manages reference data.

Analyst Service

The Analyst Service is an application service that runs the Analyst tool in the Informatica domain. The Analyst Service manages the connection between the service components and the users who log in to Analyst tool.

For more information, see the Informatica Application Service Guide.

Domain Repositories

Informatica repositories, posted on MS SQL Server databases, store metadata about domain objects. Informatica domain includes the following repository:

Domain configuration repository

The domain configuration repository stores configuration metadata about the Informatica domain. It also stores user privileges and permissions.

Model repository

The Model repository stores metadata for projects and folders and their contents, including all repository objects such as mappings and workflows. The repository also stores rules you apply during data preparation.

Data Preparation repository

The Data Preparation repository stores worksheet metadata created when you use the Enterprise Data Preparation application to prepare data.

In addition to these domain repositories, the solution also requires a repository for Hive metadata. This repository is hosted on an SQL database. It stores Hive table metadata to enable Hadoop operations.

For more information about domain repositories, see the Informatica Application Service Guide.

Clusters

Informatica uses the following HDInsight clusters.

Embedded Hadoop cluster

An embedded Hadoop cluster that the Catalog Service uses to run metadata processing and profiling jobs.

Informatica compute cluster

A compute cluster deployed on HDInsight that Enterprise Data Preparation uses to publish prepared data to the data lake.

Informatica Clients

You can use several different clients with Enterprise Data Preparation:

Administrator tool

The Administrator tool enables you to create and administer services, connections, and other domain objects.

Analyst tool

The Analyst tool enables you to analyze, cleanse, standardize, profile, and score data in your enterprise.

Developer tool

The Developer tool enables you to create and run mappings and other objects that enable you to access, transform, and write data to targets.

Command line interface

The command line interface offers hundreds of commands to assist in administering the Informatica domain, creating and running repository objects, administering security features, and maintaining domain repositories.

Bastion server

You can optionally deploy a bastion server as an element in the solution. The bastion server is installed with the Developer tool and the command line interface clients. The bastion server acts as a firewall for access to the VPC. It can also act as a remote Windows server which you can log into to run Informatica clients.

Before You Begin

Before you launch the automated deployment on Microsoft Azure, verify the prerequisites and make the choices described in this section.

License Key Prerequisite

Verify that you have a license to deploy Enterprise Data Preparation.

You supply the license key value in the Informatica License Key parameter when you configure the deployment.

Prerequisites

Before you deploy Enterprise Data Preparation on Microsoft Azure, verify the prerequisites.

- You must have a Microsoft Azure subscription with owner role.
- You must have access and permissions to create the following resources on the Azure platform:
  - Virtual Network
  - Network security group
  - Virtual machines
- You must have a valid Enterprise Data Preparation license that you have downloaded to your local machine or a location on your network.
- You must have sufficient number of CPU cores based on the instance type in the region where you plan to deploy the Enterprise Data Preparation solution as described in the following table:

Deployment Type	Minimum Virtual CPU Cores*	Maximum Virtual CPU Cores**
Small	60	78
Medium	84	102
Large	132	150

*Minimum Virtual CPU Cores: Informatica High Availability and Bastion server are disabled.

**Maximum Virtual CPU Cores: Informatica High Availability and Bastion server are enabled.

Note: Not all Azure resources are supported in all regions. See the Azure documentation to verify that the resources for your solution are supported in your desired region.

In addition to geographical regions, the solution supports government cloud regions. Contact Informatica support to check support for your desired region.

Existing HDInsight Cluster Prerequisites

Before you deploy Enterprise Data Preparation on an existing HDInsight cluster, verify the following prerequisites.

- Your password is the same for the cluster login and SSH credentials.
- The HDInsight cluster is within your deployment region.
- Use the Vnet associated with the existing HDInsight cluster.
- Verify that you have fulfilled the prerequisites for launching an HDInsight cluster. See the Informatica documentation.

Deploying Enterprise Data Preparation on the Azure Marketplace

The automated deployment of Enterprise Data Preparation on the Azure marketplace uses the Azure Resource Manager template to guide your choices and launch the solution deployment.

When you provision the solution on the Azure marketplace, launch the wizard and configure the basic properties.

Step 1. Begin Provisioning

Use the Azure Marketplace website to provision Azure cluster resources including an Enterprise Data Preparation deployment.

When you implement the solution on the Azure marketplace, you launch the wizard, and configure basic properties.

1. Search for and select the Enterprise Data Preparation solution.
   a. Log in to the Azure marketplace website. Use the search bar to search for Informatica Enterprise Data Preparation.
   b. Select **Informatica Enterprise Data Preparation 10.4.0.**

      Click **Get it now** to launch the solution wizard.
   c. Read the details of the terms of use and click **Continue**.

      The wizard redirects the browser window to the Enterprise Data Preparation 10.4.0 solution on the Azure portal.
   d. Click **Create**.

   A series of panels open to enable you to configure the solution on the Azure platform.
2. Supply information in the Basics panel, and then click **OK**.

Step 2. Deploy Enterprise Data Preparation and Configure Azure Resources

Deploy Enterprise Data Preparation and configure new or existing Azure resources to use with it.

You must download the latest mysql.jar, and then upload the archive as part of the deployment process. The file name must be `mysql-connector-java-<versiondetails>.jar`.

Basics

Enter values for the following parameters:

Parameter	Description
Subscription	Required. Azure subscription you use to manage the deployment.
Resource Group	Required. The Azure resource group containing the Virtual Network where you deploy Enterprise Data Preparation.
Region	Required. Azure location where you deploy Enterprise Data Preparation.

Informatica Enterprise Data Preparation

Enter values for the following parameters:

Parameter	Default	Description
Informatica License Key File	Requires selection	Required. Enterprise Data Preparation license key value.
Informatica Server	Standard_D5_v2	Required. Size of the Informatica server.
Database Server	Standard_D3_v2	Required. Size of the database server.
Deployment Type	Requires selection	Required. Select the cluster type for the embedded Hadoop cluster and the Informatica compute cluster. - Small - Medium - Large
Embedded Hadoop Cluster Virtual Machine Size	Standard_D5_v2 / Standard_D4_v2	Required. Size of the virtual machine that hosts the embedded Hadoop cluster.
Informatica High Availability	Disabled	Required. Indicates whether you want to enable high availability for the deployment.
Password	Requires input	Required. The password for SSH or RDP for all virtual machines, the Informatica domain, and all databases.
Confirm Password	Requires input	Required. Confirms the password that you entered.

Existing HDInsight Cluster Settings

Enter values for the following parameters to deploy Enterprise Data Preparation on an existing HDInsight cluster:

Parameter	Default	Description
Microsoft HDInsight Cluster URL	Requires input	Required. URL for the existing HDInsight cluster. Specify the full URL including `https://`. To get the URL, complete the following steps in the Ambari Web UI: 1. Select **All Resources**. 2. Filter the list based on HDInsight cluster. 3. Click on the HDInsight cluster, and then select the URL.
Azure Data Lake Storage Gen2 Account Name	Requires input	Required. Azure Data Lake Storage Gen2 account to use as the primary storage type for the HDInsight cluster.
Azure Data Lake Storage Gen2 Account Resource Group	Requires input	Required. Resource group to which the Gen2 storage account belongs.
Cluster login user name	Requires input	Required. User name to use to log in to the HDInsight Cluster Web UI.
Password	Requires input	Required. Password to use to log in to the HDInsight Cluster Web UI and SSH.
Cluster secure shell (SSH) user name	Requires input	Required. User name to use to log in via SSH.

New HDInsight Cluster Settings

Enter values for the following parameters to deploy Enterprise Data Preparation on a new HDInsight cluster:

Parameter	Default	Description
HDInsight Virtual Machine Size	Standard_D13_v2	Required. Size of the nodes in the HDInsight cluster.

Bastion Server

Enter values for the following parameters:

Parameter	Default	Description
Deploy Bastion Server	No	Optional. Deploys a bastion server to access other resources in the Virtual Network.
Bastion server size	Standard_D2_v2	Required if you choose to deploy a bastion server. Virtual machine size of the bastion server.

Infrastructure Settings

Enter values for the following parameters:

Parameter	Default	Description
CIDR IP Address Range	Requires input	Required. Classless Inter-Domain Routing IP address range permitted to access Enterprise Data Preparation. Format the string as `x.x.x.x/x`.
Assign Public IP Address	Yes	Required. Assigns a public IP address to the network interface that is attached to the virtual machine.
Virtual Network	New	Required. Select to create a Virtual Network where you deploy Enterprise Data Preparation.
Subnet	Requires selection	Required. The identifier for the subnet within the Virtual Network where you deploy Enterprise Data Preparation.
Upload MySQL JAR	Requires selection	Required. Path to the mysql.jar for the MySQL database. The file name must be `mysql-connector-java-<versiondetails>.jar`. The .jar version must be equal to or higher than 5.1.28. For more information about MySQL database requirements, see the Informatica documentation.

After you configure the parameters, verify the choices in **Review + create**, read the terms of use, and click **Create**.

When you click **Create**, Azure deploys Enterprise Data Preparation and creates resources in the environment that you configured.

Step 3. Recycle the Informatica Services

After the deployment completes, use the Administrator tool to recycle the Informatica application services.

1. Log in to the Administrator tool.
2. Click the **Services and Nodes** tab.
3. In the Domain Navigator, select the Model Repository Service.
4. From the **Actions** menu, select **Search Index > Re-index**.
5. From the **Search Index > Re-index** menu, select **Recycle Service**.
6. Select the Data Integration Service, and then select **Actions > Recycle Service**.
7. Select the Content Management Service, and then select **Actions > Recycle Service**.
8. Select the Interactive Data Preparation Service, and then select **Actions > Recycle Service**.
9. Select the Enterprise Data Preparation Service, and then select **Actions > Recycle Service**.
10. Log in to the Enterprise Data Preparation application.
11. Log out of the Enterprise Data Preparation application, and then log in again.

Monitoring Instance Provision and Informatica Domain Creation

You can use cloud platform dashboards, logs, or other artifacts to see whether cluster creation succeeded and how to locate and identify the Informatica domain on the cloud platform.

During Deployment

After you finish configuring the solution and start the deployment process, the Azure dashboard indicates deployment status in the top right corner.

To view the detailed status of the deployment job, including resources, click **Deployment in progress...**

When Deployment is Complete

The automated deployment includes the following resources:

- A Virtual Network.
- A network security group.
- A Virtual Machine hosting a Microsoft SQL Server database.
- An Informatica domain.
- An HDInsight cluster.
- An embedded Hadoop cluster.
- A MariaDB database.

Perform these steps to use your Azure dashboard to verify the status of resource deployment:

1. Use the dashboard search bar to search for the resource group that contains the Enterprise Data Preparation deployment.
   The dashboard displays the **Overview** view of the resource group, with resource deployment status as a clickable link in the upper right corner.
2. Click the resource deployment status link. A detail window opens listing the failed and successful deployments.
3. Click **Error details** for information about failed resource deployments.
4. Click **Overview** to see a list of the resources in a resource group.
5. You can click column headings in the display to sort by name, type, or location of the resource.

Output Tab

Use the dashboard search bar to search for the resource group that contains the Enterprise Data Preparation deployment. The dashboard displays the Overview view of the resource group, with resource deployment status as a clickable link in the upper right corner.

If the deployment succeeds, click the deployment named `maintemplate>Outputs`.

The following table describes the output properties:

Output	Description
username	User name of the Administrator tool user.
password	Password for the Administrator tool user.
databaseUsers	User IDs for the database users of the domain databases created in the deployment.
informaticaAdminConsole_PrivateIP	IP address for the node where the Administrator tool runs.

Output	Description
informaticaAdminConsole_DNS	DNS for the node where the Administrator tool runs.
informaticaPreparation_PrivateIP	IP address for the node where Enterprise Data Preparation runs.

Use the Administrator tool to perform administrator tasks. For a description of domain and Enterprise Data Preparation administrator tasks, see the Informatica Administrator Guide and the Informatica Enterprise Data Preparation Administrator Guide.

Logs

After the Enterprise Data Preparation deployment completes, review the log files to see the results.

You can access the following logs on the Azure Virtual Machine that hosts the Informatica domain:

Azure extension operation logs

Records the installation of Azure resources and services.

File name: `stdout` and `stderr`

Location: `/var/lib/waagent/custom-script/download/0`

Note: A subdirectory in the above path under `/download/0/` contains `stderr` and `sdtout` logs. The directory also contains the file `silentlaunch.sh`, which contains the script that was executed to install Azure resources and services.

Command execution log

This log records the following events:

- Creation of Informatica connections, cluster configurations, and services
- Population of the data warehouse and databases

File name: `oneclicksolution.log`

Location: `/opt`

Informatica domain and services configuration log

At the top of the log file is a summary section that lists automated tasks and their status. You can view the details about each task under the summary section. If any of the tasks failed complete successfully, you can look at the detailed section for the task to troubleshoot the task.

Author

Dan Hynes